



RDPMS System Requirements Specification REV 1.0

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1. Introduction

1.1. Purpose

This document defines the system and module requirement specification for monitoring the signaling equipment's data.

The objective of providing Remote Diagnostic and Preventive Maintenance system is to assist maintenance team in taking appropriate maintenance action in advance to prevent the failure of signalling gears. At functional level the system is broadly divided into two categories:

- A.** Remote Diagnostic: of equipment to reduce MTTR and providing aid to maintenance staff in rectifying the signal failures. This is achieved by automated analysis of parameters of Signal gears collected remotely using IoT devices. The system shall be able to give probable cause of failure to aid the maintenance staff for early restoration reducing MTTR.
- B.** Predictive Maintenance: Facilitating predictive maintenance by advance computing of big data using Machine Learning and Artificial Intelligence. The data of all the stations shall be continuously analyzed by system for developing the supervised and unsupervised machine learning. The system shall be able to send automatic alerts for Signalling gears which are likely to fail based on the system learning. This will assist the maintenance staff to take necessary action to eliminate failure before it occurs.

1.2. Scope

This document contains the functional requirement for Remote Diagnostic and Predictive Maintenance System for Indian Railway Signaling equipment using IoT based data acquisition systems and Machine learning/Artificial Intelligence.

So, the specification can be divided into system requirements and module requirements.

To ensure that the Specification is not restrictive to propriety systems, the functional requirements have been laid down and device specific details are avoided to keep scope of widely available systems without any propriety. hardware and protocols. Hence, the scope of this specification is to standardize the requirements for ensuring conformity and interoperability such that data being collected on any platform/IoT device shall be utilized for machine learning/AI based data processing in the system.

1.3. Definitions, Acronyms, and Abbreviations

IoT	Internet of Things
AI	Artificial Intelligence

ML	Machine Learning
I/O	Input/output
MTBWSF	Mean Time Between Wrong Side Failure
MTTR	Mean Time to Repair
MTBWSF	Mean Time Between Wrong Side Failure
MTBF	Mean Time Between Failure
PCB	Printed Circuit Board
EI	Electronic Interlocking
OFC	Optical Fiber Cable
IPS	Integrated Power Supply
RTC	Real Time Clock
LoRa	Long Range communication (a wireless Technology)

Table 1: Abbreviations

1.4. Applicable Documents for reference

Si. No	Subject	Specification
1	IEC 60688	Electrical measuring transducers for converting AC and DC electrical quantities to analogue or digital signals
2	IEC 61326	Electrical equipment for measurement, control and laboratory use – EMC requirements –
3	IEC 61754-20	Fiber optic interconnecting devices and passive components – Fiber optic connector interfaces – Part 20: Type LC connector family
4	IEC 60603-7	Connectors for electronic equipment – Part 7-7: Detail specification for 8-way, shielded, free and fixed connectors, for data transmissions with frequencies up to 600 MHz

5	EN 50121-2/ IEC	Railway applications – electromagnetic
6	IEC 62236-2	Compatibility – Part 2: emission of the whole railway system to the outside world
7	EULYNX	EULYNX EULYNX standards : selective Docs.for
8	IRS: S-24	Electric Point Machine
9	RDSO SPN 153	Subsidiary LED Signal
10	RDSO SPN 199	Main LED Signal
11	RDSO SPN 146	Audio Frequency Track Circuit
12	IRS S 105	Block Proving by Axle Counter using UFSBI
13	RDSO SPN 175	Solid State Block Proving by Digital Axle Counter
14	RDSO SPN 176	Multi-section Digital Axle Counter
15	RDSO/SPN/177	Single section Digital Axle Counter
16	IRS: S-99	Data Logger System.
17	IRS: :S 63	PVC insulated underground unscreened cable for Railway signaling
18	IRS: S 76	PVC insulated indoor cables for Railway signaling
19	RDSO/SPN/144	Safety and reliability requirement of electronic signalling equipment.
20	RDSO/SPN/197	Code of Practice for Earthing and Bonding System for Signaling Equipment's.
21	EN 50121*	Railway Applications - Characteristics of Railway Systems that affect EMC behavior
22	ISO 9001	Quality Systems- model for quality assurance in design, development, production, installation and serving.
23	IEC 60688	Electrical measuring transducers for converting. a.c. and d.c. electrical quantities to analogue or digital signals.
24	IEC61326	Electrical equipment for measurement, control and laboratory use –EMC requirements

Table 2 : Reference Documents

SYSTEM REQUIREMENTS SPECIFICATION

2. SYSTEM ARCHITECTURE:

The system consists of the following:

- 2.1** Data acquisition for parameters of Signaling equipment's using additional sensors and/or inbuilt diagnostic ports of the equipment.
- 2.2** Transfer of raw data from site using IoT devices to local server at station.
- 2.3** The local server is complemented by Edge computing device to generate first level fault and diagnostic information based on the real time data.

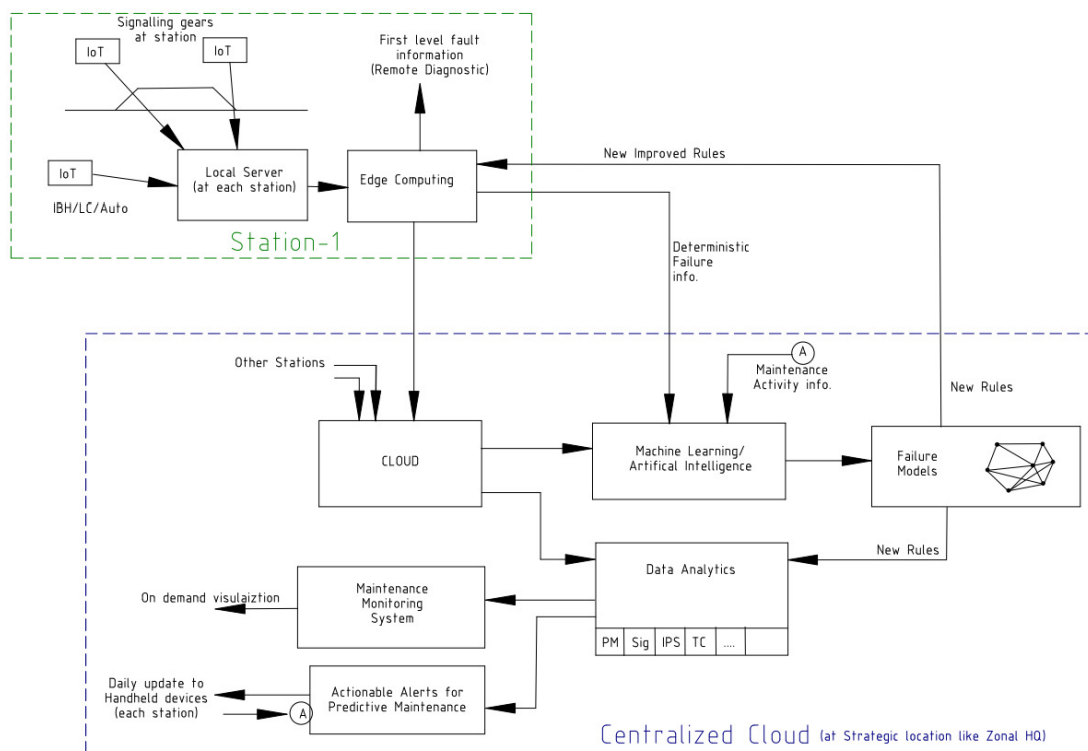


Figure-1 : System Architecture

- 2.4** The data from local servers at every station will be sent to centralized cloud. This data complemented by historical data and trends will be processed using advance computing through data analytics (Artificial Intelligence / Machine Learning) to anticipate any issues or failures before they occur.
- 2.5** Remote monitoring: An automated alert mechanism will send information to Signal Control of concerned Division and the concerned maintenance staff over sms/mobile app.
- 2.6** Predictive Maintenance: Mobile app should also be developed which shall be having real time update on each station gears. The maintenance staff of particular station will be having the handheld device where at the start of each day he can see which gears need the specific

attention for maintenance.

3. DATA ACQUISITION:

3.1 The data from each signaling gear at the station shall be collected on real time basis using the inbuilt diagnostic ports of the signaling gears and/or the external sensors.

3.2 General requirements for external sensors:

For measuring the currents non-intrusive sensor shall be used. Voltage sensor shall also be preferably non-intrusive. In case of non-suitability/non availability of non-intrusive sensor for voltage transducer having high impedance and galvanic isolation can also be considered with minimum withstand voltage of 2.5KV for 60 sec. Loading on signaling circuit should be minimum and in no case it should be more than 5ma.

3.2.1 The accuracy of sensors should be such that it will not cause noise in the data being used for machine learning. Proper accuracy and stability of sensors is very important for end goal of predictive maintenance. If the raw data is noisy and not reflecting true value of field gear, the machine learning will be inaccurate.

3.2.2 Most of the sensors would be transducers converting current, voltage, vibration, etc to very weak electrical signals. Hence, proper safeguard against EMI effects on the stability and accuracy of readings should be taken. Protection arrangement for sensors against surges coming from lightning/power line/ OHE shall be made. Appropriate signal processing technique and filtering shall be used to remove unwanted signals.

For example, in the RE area, the induced voltage is dynamic varying due to change in OHE current during presence or absence of train. Hence, the length of cables/wires from transducer/sensor to IoT device shall be such that induced voltage are within limit to ensure desired accuracy of stable readings.

Further, the transducer/sensor wires should not run parallel to dynamic current carrying conductors which may alter the readings of the sensors.

3.3 Proper rated sensors shall be used as per parameter monitoring range of gear type to be monitored. Since the machine learning will depend on the variation of data to capture health signature, the least count of sensor for each category of equipment would be different. Sensor should comply to IEC 60688 and accuracy shall be of class 1 as per table 1 of IEC 60688 and environmental condition shall be of usage group-II (K 70) as per table 5 of IEC 60688. Sensor should also comply immunity requirement as per table-2: immunity test requirement for equipment intended to be used in an industrial electromagnetic environment, of IEC 61326. Test certificate in this regard shall be submitted. Provision of on-site calibration every year by using suitable measuring equipment duly calibrated by NABL accredited Lab shall be kept.

For critical faults, the sms or automated voice calls may be provided once machine learning is stable and alerts are quite accurate (upto > 60%).

The system architecture is depicted broadly in the Figure-1. As Signaling assets are distributed at more than 6000 stations, the decentralization of computing is necessary to handle the requirement for which edge computing at every station is proposed. However, the machine learning need to be centralized for complete Indian Railways so as to use the data from every station as well as streamlining the efforts for machine learning techniques. Hence, the need for one platform is proposed for AI techniques.

OUTDOOR GEARS AND THEIR TENTATIVE PARAMETERS TO BE MONITORED

A. Signals HMU:

The following data shall be recorded for each aspect of Signal:

- i) AC Current.
- ii) AC Voltage.
- iii) Status of local signal control/detection relays at Location boxes.

B. Points HMU:

The following data shall be recorded for each end of the Point:

- i) DC operating Current.
- ii) DC operating Voltage.
- iii) Vibration.

C. TCK HMU:

The following data shall be recorded for each Track Circuit:

- i) DC feed end and relay end voltage.
- ii) DC feed end and relay end current
- iii) Input Voltage & output current of Track feed battery charger,
- iv) Voltage & charging /discharging current of battery.
- v) Incoming & outgoing TPR voltage.

D. Presently the external sensors are required to be provided for monitoring the health of Points, Signals and DC Track circuits as these are yet not inbuilt on Indian Railways. The details of nominal voltage/current range for the Signal, Points and DC track circuits and other information is given in Annexure-1 for ready reference.

E. Axle Counters (SSDAC, MSDAC etc):

Input Voltage, current and status of vital relays and Diagnostic data from Diagnostic port of Axle counter evaluator shall be captured. Protocol shall be shared by OEM for sharing diagnostic data.”

F. AFTC: Voltage, current, frequency data of EJB installed in field and Diagnostic data from Diagnostic port of AFTC evaluator shall be captured. Protocol shall be shared by OEM for sharing diagnostic data.”

G. Level Crossing Gates:

The following data shall be recorded for each Boom:

1. AC/DC operating Current of Electrical Lifting Barrier.
2. AC/DC operating Voltage of Electrical Lifting Barrier.
3. Status of open close position for each boom.
4. Locking status of gate.

Note*: Voltage and current signature of motor shall be recorded by capturing data at the rate of 20ms during operation of ELB.

INDOOR Gears AND THEIR TENTATIVE PARAMETERS TO BE MONITORED :

A. Relays:

Status (pickup or drop) of each relay in the relay room shall be recorded through datalogger.

B. Electronic Interlocking:

Data from Diagnostic port of Electronic Interlocking.

C. Integrated Power Supply (IPS) / Battery Charger: The following shall be monitored:

- i) All voltage outputs
- ii) Health of IPS modules from potential free contacts
- iii) Availability of standby input power supplies from Auto-changeover system of IPS.

D. Surge Protection Devices:

The potential free contacts of SPD devices where available shall be monitored. Care should be taken to keep sensor wiring from SPD such that (electrically) dirty cable in SPD box is not in parallel to sensor wires.

E. OFC Cable:

Monitoring of Dark Fibre for monitoring the health of OFC cable, especially those not under NMS monitoring (like OFC for distributed EI, EI to ASM room, etc). The dark fibre of the cable shall be used to detect the following:

- i) OFC cut and Geographical Location of Cut.
- ii) dB Loss / Degradation.

F. All Networking switches/dc-dc convertors and other devices having potential free contacts for monitoring.

G. Fuse Auto changeover system: Status of main and standby fuse.

- H.** Battery: Overall Voltage and current of 110V battery bank will be monitored. The health of battery to be monitored by using time proven techniques available in other industries using IoT. – Industry experts for suggestion and comments

Note: The IPS battery set consists of 55 Numbers of individual 2V Lead Acid cells connected in series for 110V DC backup. Whether monitoring of each cell is recommended or there is any established technique to monitor health using overall Voltage/Current. Any other established technique (which does not require to monitor each cell) available for monitoring of battery health may also be deployed. (comment regarding availability of such system may be provided)

- I.** Earth Leakage Detector (ELD):

Status of earth leakage from the ELD to be monitored.

- J.** Cables:

The use of coordinated data shall be made by the machine learning to monitor the voltage drop analysis for checking the health of signaling and power cables. Hence, existing data collected from voltage in IPS room, Voltage in field, ELD data etc will serve for the health of cables.

- K.** Block Instruments:

Line current and selective data for different types of Block Instruments using non-intrusive sensors.

- L.** UFSBI:

Relay status and Potential free contacts of UFSBI modem, Data from Diagnostic port, etc shall be monitored.

Note: The information already being logged in conventional Data logger shall be used.

- M.** Ambient Temperature and humidity of Battery room, relay room, power room, IoT devices location, etc shall be recorded. The temperature data may help in machine learning algorithms to model the environmental effects.

- N.** Any other equipment not in above list or any additional parameter of above listed equipment as found suitable for better predictive maintenance will be incorporated. However, to avoid unnecessary wiring for monitoring, it is recommendatory to monitor only required gears and parameters

- O.** Due to issue of proprietary protocols in signaling equipment's, which is causing issues in interfacing various electronic signaling equipment's, Indian Railways is working for

uniform protocols like EULYNX.

In this regard following will be followed:

OEM of equipment will provide protocol of his system. In future only EULYNX std shall be followed by all OEM.

OEM of RMPM system must follow EULYNX standard.

IoT device:

A. The IoT device shall be software embedded system **preferably COTS (commercially off the shelf)** and will do the basic function of capturing the parameters from the Signaling devices using the sensors and diagnostic ports and transmit the data to Local Server through Gateway at the station.

B. For monitoring at IBS/LC/Auto Signalling gears, the IoT device shall be linked to nearest station and its local server.

C. It shall work on available 24V DC power of IPS as available at site.

D. To ensure that the interoperability between different manufacturers, the standard protocols for communication may be used. The protocols of EULYNX standard for Diagnostic information of Signaling equipment shall be followed.

E. It is imperative that the IoT device will continuously monitor the sensors for the parameters depending on scanning interval. Scanning interval to be defined keeping in mind requirement for machine learning & AI and availability of COTS IOT devices. However, device should support scanning interval of 20ms or less, it should be possible to program scanning interval.

F. Further, the data captured by IoT device will be transferred to Local Server. However, if the data is transferred every cycle, there would be two issues: first is the bandwidth requirement for channel and second is the large data (50 packets per second per parameter) which may be difficult to handle and will be ignored at processing level at higher stages. Hence, selective methodology is required to be made for each type of device as per general guidelines below:

Some examples for ready reference are as below:

- i) Track circuits and Signal Lamp which operates continuously, the operational parameters may be sent conditionally whenever change of reading is more than specific percentage (say 2%) of last sent value.
- ii) However for Point machine & ELB motor which operates only for few seconds when operated, the operating current for every scanning cycle will be important to capture its **current signature**. The current signature may help in deciding machine learning to predict cases of less lubrication, friction clutch issue, stone obstruction and gap, Health of motor etc. For such application data will be captured & stored at an interval of 20ms.

- G. Accordingly, the methodology for sending data from IoT device to server will be laid down for each equipment. **The IoT being embedded system shall be customizable to get the rules changed by the firmware update remotely.**
- H. For diagnostic purpose provision shall be made that if any sensor or embedded device in field goes down sms alert may be generated. Log of the same shall also be recorded at station server.
- I. IoT device should be rugged and industrial grade suitable for 24x7 working.
- J. IOT devices shall be tested as per RDSO/SPN/144 from Signal Lab RDSO or any other NABL accredited government Lab. IOT device should comply immunity requirement as per table-2: immunity test requirement for equipment intended to be used in an industrial electromagnetic environment, of IEC 61326. The test certificate shall be submitted.
- K. System should be modular and scalable in design to make it economical, preferably rack mountable. It should be compact in nature so as to be accommodated in existing location boxes.

Network and Communication protocols between subsystems:

- A. The exchange of information from IoT device to Local server and to Centralized Cloud shall be on standard format for interoperability between different systems.
- B. The network architecture and communication protocols for Diagnostic data as defined in EULYNX standards shall be followed. These are covered in Annexure-C.
- C. It is planned for adoption of standard protocols for all Signaling equipment in future. Hence, system need to be compatible for direct interfacing with the diagnostic ports of various equipment like Electronic Interlocking, Axle counters, etc.
- D. As defined in EULYNX, the **protocols OPC-UA** shall be used to transfer diagnostic data from the connected systems (IoT) to the service function Diagnostics collector (Local server/Edge device).
- E. The time synchronization is an important requirement between various IoT devices for data interpretation in machine learning. This has been taken care in the standard protocol. For master clock, the IRNSS (Indian Regional Navigation Satellite System) clock or SNTP protocol shall be taken for reference.
- F. The layers of network architecture as per EULYNX standard for the complete Diagnostic system is as shown in figure,

Application Layer = OPC-UA	Application layer =NTP
Transport Layer =UDP/TCP	Transport Layer =UDP
Network Layer	
Data Link Layer	
Physical Layer	

- G.** At Physical Layer suitable network media shall be used. The EULYNX standard requires OFC and RJ45 (Copper) media which runs along with the equipment. However, on Indian Railways the field devices not having inbuilt diagnostic port shall be connected using IoT devices for which option of wireless media is preferable.
- H.** The following may be the media between various devices.
 For communication from IoT device (in station yard near point, signals and other equipment's) to Gateway at station—
- i) Wireless media on LoRA, Zigbee or any open source technology in free band of wireless spectrum. In case of connectivity issue on the open source free band wireless system due to higher distance or due to other technical reason, 4G/LTE may be used for that location.
 - ii) Optical fiber LC interface [IEC 61754-20] (mandatory for outdoor field IoTs)
 - iii) Copper cable. (RJ45 interfaces [IEC 60603-7])
 - iv) Wi-Fi communication.
- I.** For communication from outdoor type IoT devices (for field gears like signal, point, track circuits, etc) to Gateway at station, the wireless connectivity is mandatory along with optional Fibre connectivity (Type LC connector).
- J.** While for communication from indoor type IoT device (for IPS, Axle counter evaluator, Datalogger, etc) to Gateway at station – RJ45 (copper) and/or LC port (Fiber) may be used. Redundant communication path (wired/wireless) must be kept for reliability.
- K.** For far away field IoT's like for IBH, LC gates wired media like OFC or wireless system like 4G/LTE may be used.
- L.** For communication from Station to Cloud, existing optical fibre network of Railtel may be used. For redundancy 4G/LTE may be used along with OFC network.
- M.** The details of higher network layers, data packet structure, etc are defined in Annexure-C which has been taken from EULYNX. In this connection, the firm wares of IoT devices and other backend software of system shall be designed in such flexible manner to carry out minor update on change of protocol whenever required in developmental stage.

Local Server and Edge computing:

- A.** At each station there shall be centralized setup consisting of Gateway to collect the

data from various IoT devices and a set of Local server with Edgecomputing. One integrated unit with capability to work as gateway to collect field data ,edge computing etc can also be used. The basic architecture at station is depicted in Figure-3.

- B. The Gateway shall have interface to collect data from various media like wireless, OFC and copper. It shall also be able to collect data from conventional Data Logger for indoor relays status.
- C. The edge computing shall be employed to decentralize the computing and handling the big data and alert mechanism for each station.
- D. Edge computing shall be done by suitable PC based system. **Industrial grade hardware shall be used for 24x7 operation unmanned functioning.**
- E. The following is the broad functions of Edge computing device:
- F. Scrutinizing the local data of signaling gears to generate real time alerts on the “deterministic failures”.

For example – A Point has no indication of Normal or reverse for morethan 10 second. The system will detect this as failure and using the data of voltage/relay monitoring at relay room, point motor, etc, it should generate analarm that “Point

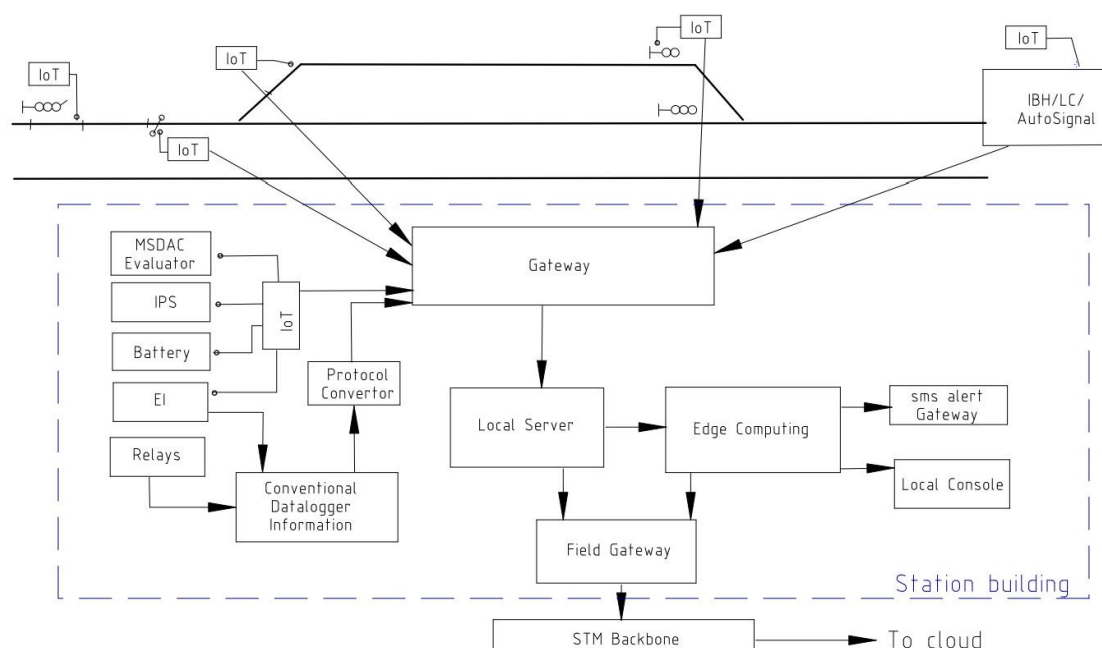


Figure-3 : Arrangement at Station

No. 101 of Station ‘ABC’ has failed. Probable reason is – detection contact problem at A end”

There shall be equipment specific deterministic failure patterns which shall be evolved during development. Systematic and Formal approach is required for writing these rules.

- G. Sending the alert to maintenance staff through sms using “SMS Alert Gateway” over GPRS/LTE.
- H. This alert shall also be sent to Cloud which shall be used for improving machine learning algorithms using AI techniques.
- I. The edge computing device will update itself by new rules of deterministic failures detection sent by the Cloud based on Machine learning.
- J. The Local server at each station of suitable size shall be provided. The detail requirement would be evolved during development.
- K. There shall be a local monitoring maintenance system at each station which shall display real time data and on demand information from local server.
- L. The complete setup for Local server and edge computing shall be provided Relay room or Maintainer room or any suitable location at the station. However, the local monitoring maintenance system shall be provided in Maintainer room.

Field Gateway:

- A. At each station there shall be field Gateway which filters and pre-processes the data to be sent to and received from Cloud.
- B. Field gateway will transmit/receive the data to cloud over the STM backbone of Railtel. 4G/LTE communication may also be used for redundancy.

Annexure-D

(Informative)

- D.1 Failure model examples:
- D.2 The point's operating current signature (variation of current value with time during operation) can differentiate between various reasons of failures. For example, point failed due to stone obstruction between tongue rail and stock rail will have sudden rise of current after smooth displacement. While it will be different current signature for poor lubrication.
- D.3 Similarly, based on particular point's historical data the system can know the displacement of tongue rail. The system can predict if the point has failed due to some obstruction (incomplete movement) by seeing the time length of normal current and point where current increase (obstruction). If current dropped at average time, then movement of tongue rail is full which shows that failure is possibly due to other cause like detection contact failure.
- D.4 The health of cable core can be accessed by measuring the voltage at relay room and the voltage reaching the point machine. The drop of large voltage during particular point operation can be used to detect faulty cores in cable.

- D.5 The DC track circuit failure identification in case of high leakage current can be predicted based on history of relay end current. If the leakage has increased gradually, it shows the poor ballast resistance due to rain or water-logging. While a sudden leakage current may show sleeper shorting.
- D.6 The total number of operations of a relay may be used to assess its life. Automated learning through AI techniques may find a trend in the life of relay in terms of number of operations and duration of cumulative pickup time of re-lays. Environmental parameters like temperature and humidity may have a relation to failures.
- D.7 In case of cable cut, the disconnection is preceded by sudden break and make of cable continuity. The voltage/current monitoring signature of different cores of cable before discontinuity can be used to identify such incident. While sudden break of continuity may imply fuse blown.

SIGNAL HMU REQUIREMENT SPECIFICATION

Signal HMU is the module responsible for sensing the signal current and voltages using the copper wires which is located either in the location box or at their respective relay cabins.

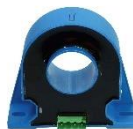
Below are the following parameters of the signals which need to be monitored,

SIGNAL ASPECTS PARAMETERS				
SI.NO	TYPE OF GEARS	TYPE OF SPECS	DESCRIPTION	PARAMETER
1	Signal Aspects	Main Signals	Rated voltage at Input terminals of Current Regulator	110V AC +25%
2			Current at rated voltage per unit at Input terminals of Current Regulator	140 mA +10%,-20% (rms)
3		Calling on Signal	Rated voltage at Input terminals of Current Regulator	110V AC $\pm$ 20%
4			Current at rated voltage per unit at Input terminals of Current Regulator	150 mA +10%, -20% (rms)
5		Route Lighting unit	Rated voltage at Input terminals of Current Regulator	110V AC $\pm$ 20%
6			Current at rated voltage per unit at Input terminals of Current Regulator	25 mA $\pm$ 5%(rms)
7		Shunt Lighting unit	Rated voltage at Input terminals of Current Regulator	110V AC $\pm$ 20%
8			Current at rated voltage per unit at Input terminals of Current Regulator	55 mA $\pm$ 5%(rms)

The above parameters should be monitored from individual signals and the following precautions should be taken for the design on signal HMU,

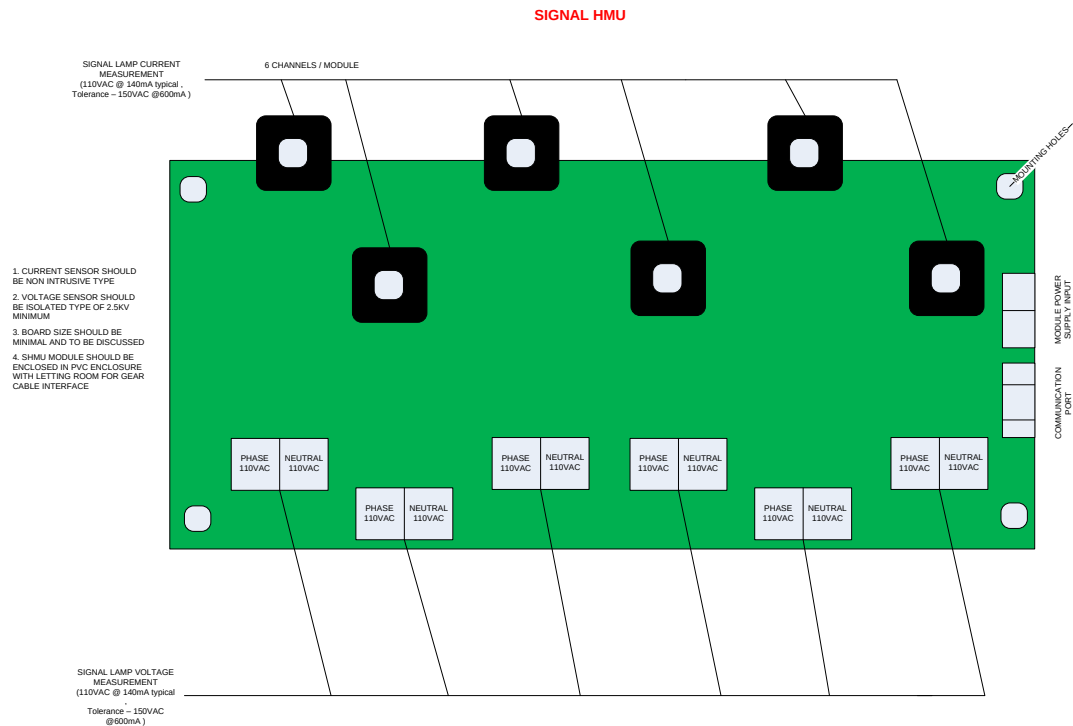
Functional Requirements:

1. Signal current should be monitored in the form of non- intrusive type.



2. Signal voltage should be monitored in the form of Galvanically isolated type of about 2.5kv minimum.

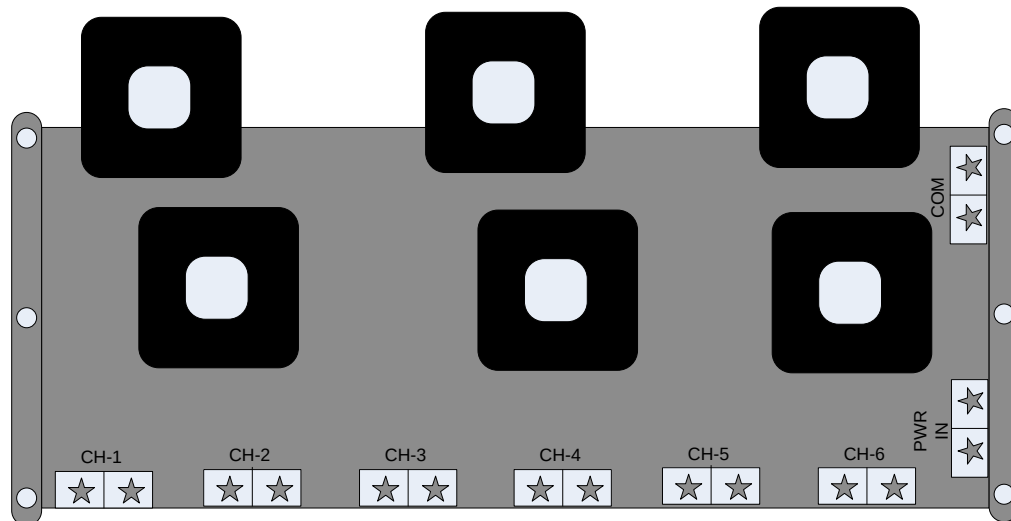
3. Short circuit, over voltage protection should be given into the circuit for additional safety.
4. Each signal HMU module should contains the minimum of 6 No's current sensors and 6 No's voltage sensors. (During design process the variation of the channels can be discussed)



5. Communication for the signal HMU to the main modules can be SM BUS, CAN BUS protocols.
6. Module should be able to work under the temperature from -40 to 85 Degree Celsius minimum.
7. Accuracy of the module data should be 100%. Tolerance level can be at 1%.
8. Module working voltages should be of 17volts – 36 Volts DC.

Mechanical Requirements:

1. Signal HMU should be able to enclose in the molded enclosure which can be specially designed or any other matching.



2. It should be able to mount the module both in Hylem sheets and C-channels.

POINT HMU REQUIREMENT SPECIFICATION

POINT HMU is the module responsible for sensing the point machines current, voltages and the vibrations using the copper core wires and the sensor wires which is located either in the location box or at their respective relay cabins. Below are the following parameters of the signals which need to be monitored,

POINT MACHINES PARAMETERS					
SI.NO	TYPE OF GEARS	THROW OF POINT MACHINES	DESCRIPTION	NOMINAL PARAMETER	SENSOR PARAMETER
1	Electric point machine	143 mm +/- 2	DC VOLTAGE	110V DC	0- 170V DC
2			DC CURRENT	<= 4.5 Amp	0- 10Amps DC
3		220 mm.+4/-1	DC VOLTAGE	110V DC	0- 170V DC
4			DC CURRENT	<= 5.5 Amp	0- 10Amps DC

The above parameters should be monitored from individual points and the following precautions should be taken for the design on point HMU,

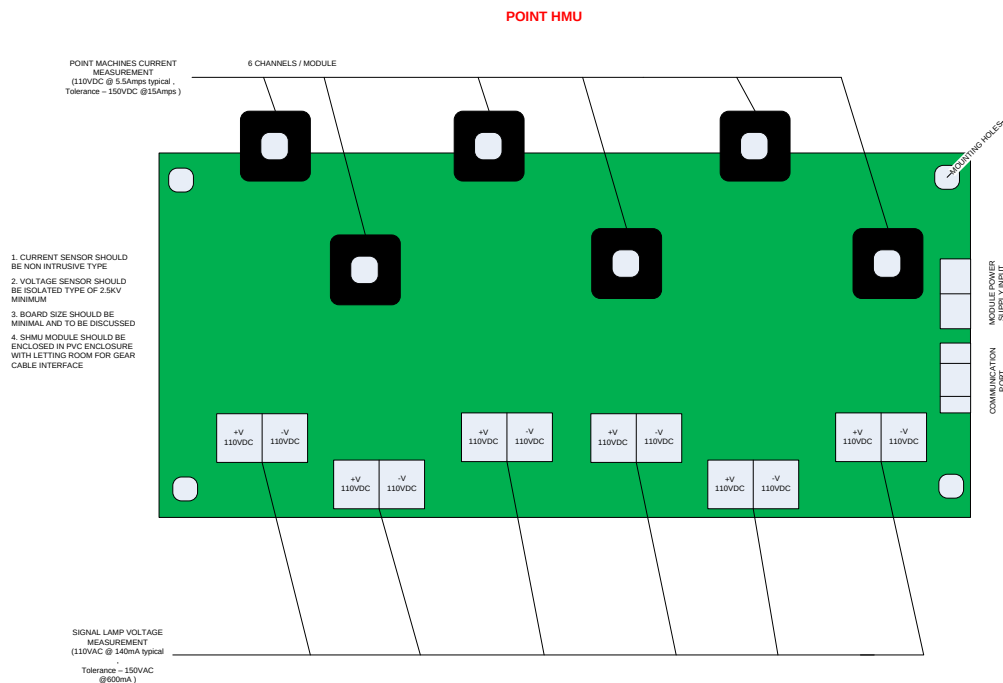
Functional Requirements:

1. Point current should be monitored in the form of non- intrusive type.



2. Point voltage should be monitored in the form of Galvanically isolated type of about 2.5kv minimum.
3. Vibration sensor should be installed inside the point machines which will be available in the field side of the each locations. So, its interface will be available in this point HMU modules for the sensor wire interface . Analog value or ADC value of the vibration sensor must transmit the data to its respective channels located in the HMU modules.
4. Short circuit, over voltage protection should be given into the circuit for additional safety.

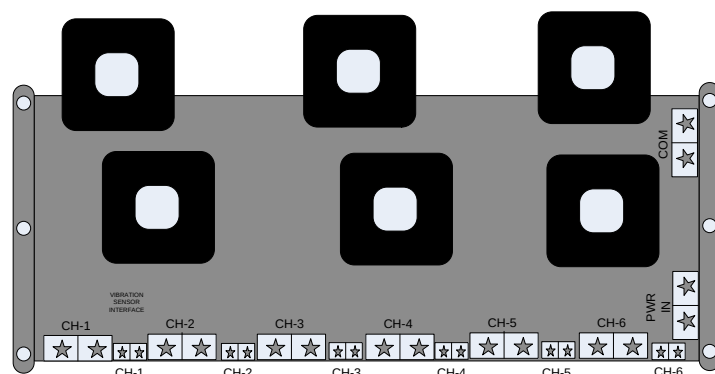
- Each Point HMU module should contains the minimum of 6 No's current sensors and 6 No's voltage sensors. (During design process the variation of the channels can be discussed)



- Communication for the Point HMU to the main modules can be SMBUS, CAN BUS protocols.
- Module should be able to work under the temperature from -40 to 85 Degree Celsius minimum.
- Accuracy of the module data should be 100%. Tolerance level can be at 1%.
- Module working voltages should be of 17volts – 36 Volts DC.

Mechanical Requirements:

- Point HMU should be able to enclose in the molded enclosure which can be specially designed or any other matching.



TCK HMU REQUIREMENT SPECIFICATION

TCK HMU is the module responsible for sensing the DC track circuits current, voltages and using the copper core wires which is located either in the location box or at their respective relay cabins. Below are the following parameters of the signals which need to be monitored,

SI.NO	TYPE	Type of Relay	Resistance of Track Relay (L= Length of the Track Circuit)	DESCRIPTION	PARAMETER	Cells at Feed end
1	DC track circuits (Non-RE area)	Non ACI shelf type	For L < 100 m → 9 Ω	PU Voltage Approx	0.4 V	1 cell (2 V)
2				PU Current Approx	40 mA	1 cell (2 V)
3		Non ACI Plug in Type (QT2)	For L > 100 m → 2.25 Ω	PU Voltage Approx	0.2 V	1 cell (2 V)
4				PU Current Approx	80 mA	2 cells (4 V)

SI.NO	TYPE	Type of Relay	Resistance of Track Relay (L= Length of the Track Circuit)	DESCRIPTION	PARAMETER	Cells at Feed end
1	DC track circuits (AC RE Area)	ACI Shelf type	9 Ω	PU Voltage Approx	0.68 V	1 cell (2 V)
2				PU Current Approx	72 mA	
3		ACI Plug in Type QTA2	9 Ω	PU Voltage Approx	1.4 V	2 cells up to < 100m 3 cells > 100 m to 450m 4 cells up to 750m
4				PU Current Approx	140 mA	
5		ACI Plug in Type QBAT	9 Ω	PU Voltage Approx	1.75 V	2 cells up to < 100m 3 cells > 100 m to 450m 4 cells up to 750m
6				PU Current Approx	175 mA	

SI.NO	Relay Style	Max P.U.Volts	Max P.U Current (MA)	Min P.U. Volts	Min. P.U . Current (MA)	Max. Permissible Volt	Min. regd. Volts	Usage area
1	QT2	1.158	117	0.834	103	300% PUV	125% PUV	in non – RE Areas.
2	QT2	0.515	117	0.37	103	300% PUV	125% PUV	in non – RE Areas.
3	QTA2	2.02	92	0.46	81	300% PUV	125% PUV	Upto 450 meters track Circuit Lengths in RE Areas.
4	QTA2	1.38	140	0.97	120	300% PUV	125% PUV	Upto 450 meters track Circuit Lengths in RE Areas.
5	QBAT	1.733	175	1.134	140	235 % PUV	122% PUV	Upto 750 meters track Circuit Lengths in RE Areas.

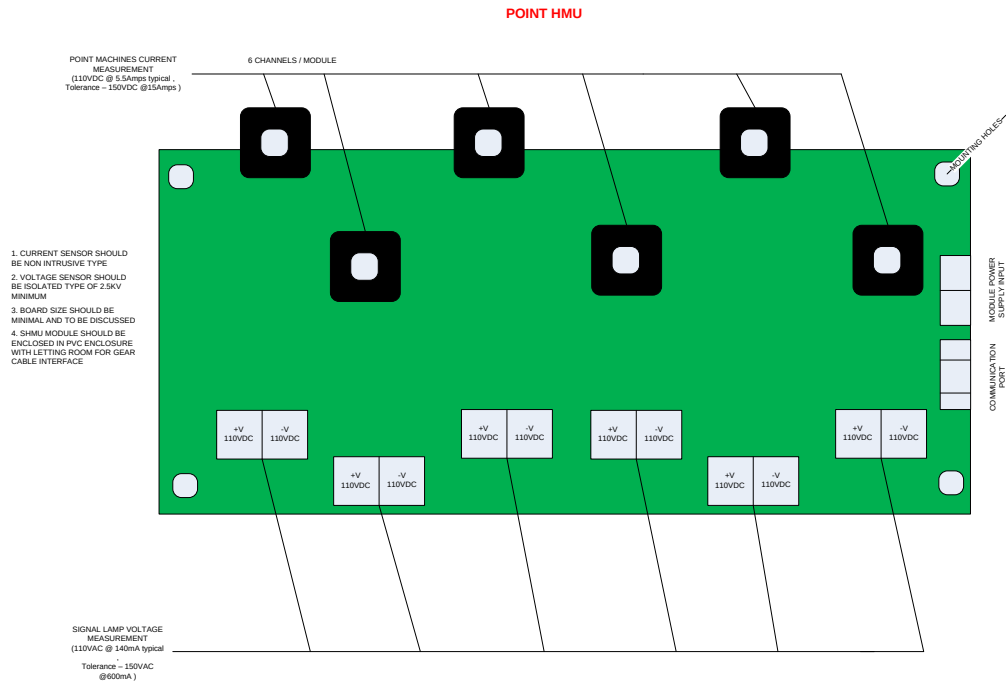
The above parameters should be monitored from individual points and the following precautions should be taken for the design on TCK HMU,

Functional Requirements:

1. TCK current should be monitored in the form of non- intrusive type.



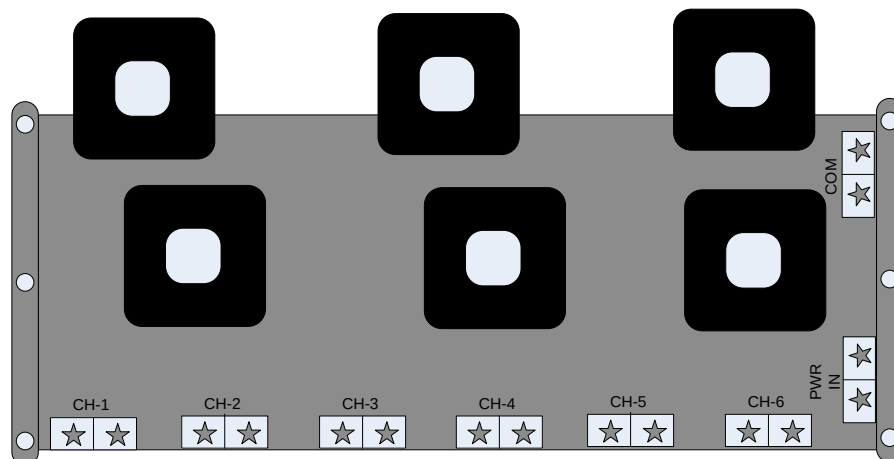
2. TCK voltage should be monitored in the form of Galvanically isolated type of about 2.5kv minimum.
3. Short circuit, over voltage protection should be given into the circuit for additional safety.
4. Each TCK HMU module should contains the minimum of 6 No's current sensors and 6 No's voltage sensors.



5. Communication for the TCK HMU to the main modules can be SMBUS, CAN BUS protocols.
6. Module should be able to work under the temperature from -40 to 85 Degree Celsius minimum.
7. Accuracy of the module data should be 100%. Tolerance level can be at 1%.
8. Module working voltages should be of 17volts – 36 Volts DC.

Mechanical Requirements:

1. TCK HMU should be able to enclose in the molded enclosure which can be specially designed or any other matching.



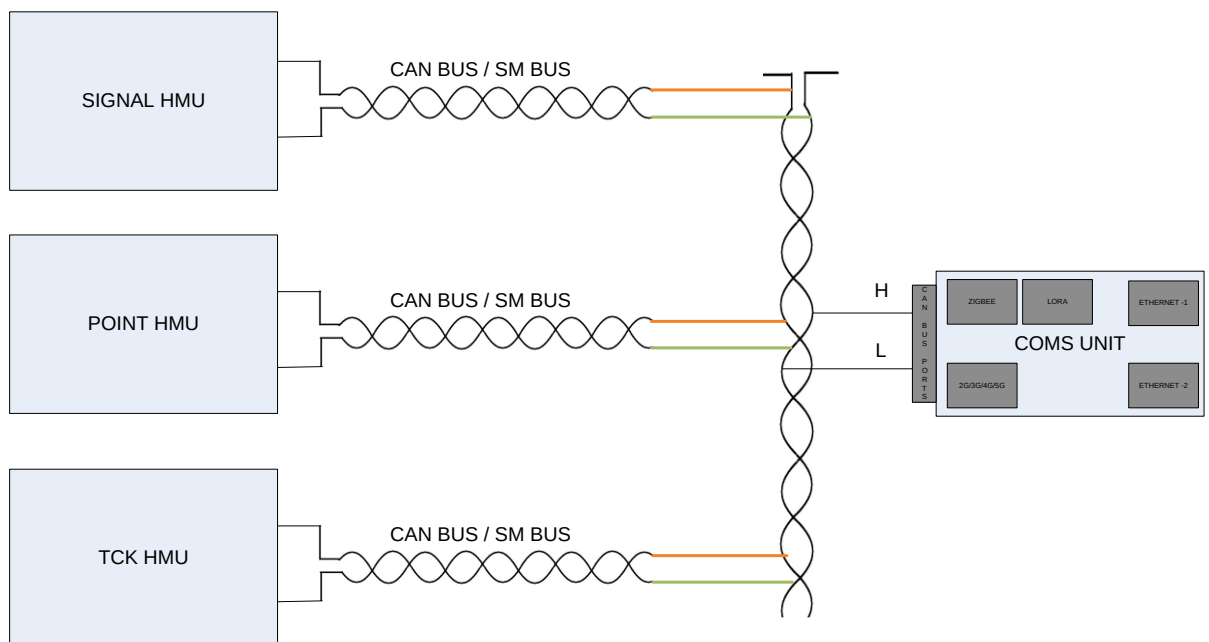
COMS UNIT REQUIREMENT SPECIFICATION

COMS unit refers to the communication module, which is responsible for collecting all field gear's data from SIGNAL HMU, POINT HMU, TCK HMU, ANALOG HMU, DIGITAL HMU, MASTER HMU of their CAN BUS or SM BUS.

This unit should contain the following communication interfaces options,

Functional Requirements:

1. Ethernet port of 2 No's minimum for external data transfer to MASTER HMU via External OFC switch needs to be implemented.



2. Zigbee communication is required for short range data transmit from location available HMU modules to relay distributed cabins coms unit over through a mesh network of intermediate devices.

Data transmission range of about 80-100 meters from location to relay cabins.

3. Lora communication is required for long range data transmit from location available HMU modules to relay distributed cabins coms unit.

Data transmission range of about 3 Kilometers from location to relay cabins either independently and through mesh network topologies.

Lora transmission frequency of india 865 MHz – 867 MHz bands, must be driven for license free data transceiver and options for customizing the frequency of license

acquiring channel bands also can be possible in futures.

4. 2G/3G/4G/5G communication is required for communicating the Master HMU modules.

Each communication mentioned can work independently or by configuration specific one.

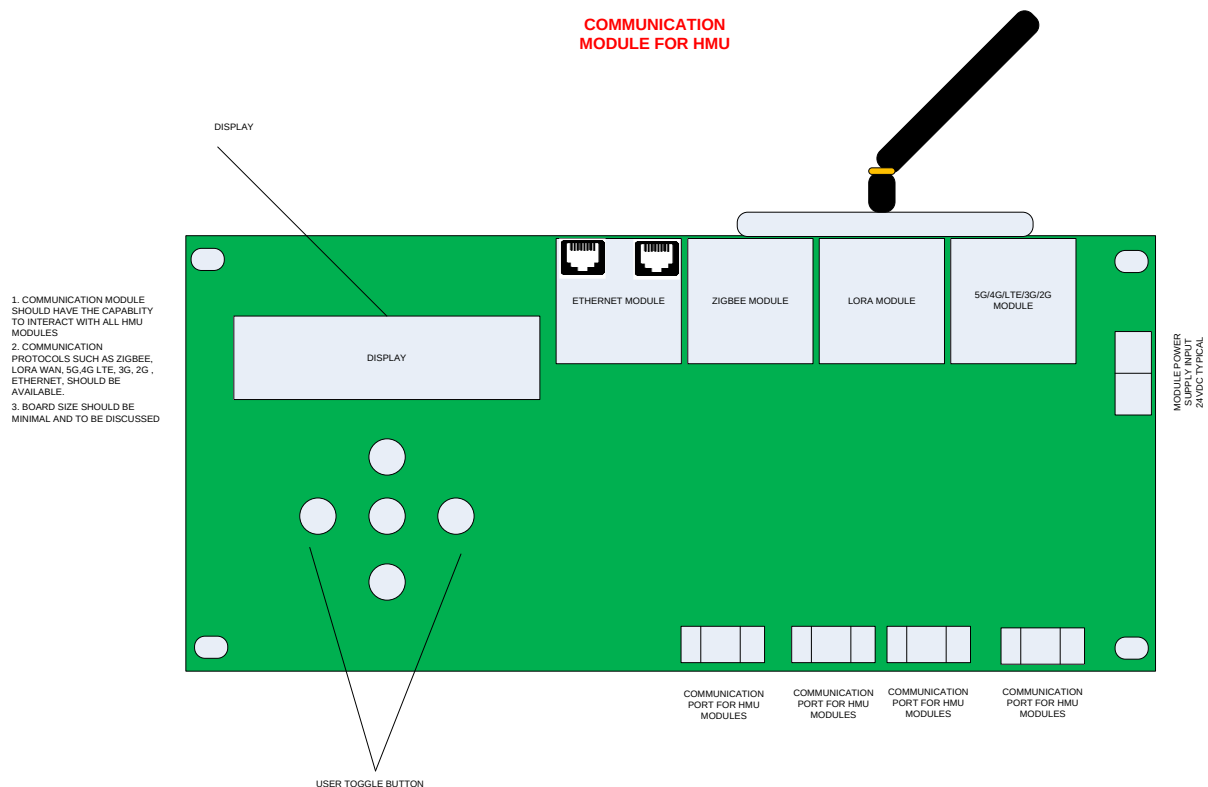
5. Wi-Fi communication can be provided as the additional data transmission medium.

2.5 GHz AND 5 GHz toggling options can be possible.

40 METERS.

6. LCD or Matrix display over 400-700 nits' brightness should be available in the coms unit along with menu push buttons for data viewing from SIGNAL HMU, POINT HMU, TCK HMU, etc.

LCD or Matrix display can be populated or non-populated depends on the requirements, so, non-populate of the display should not affect the function of the comms unit.



7. CAN or SM BUS interface needs to be given for at least 4 No's should be provided for

interfacing the other sensor modules such as SIGNAL HMU, POINT HMU, TCK HMU.

CAN BUS data rate of about 1 Mbps min should be provided.

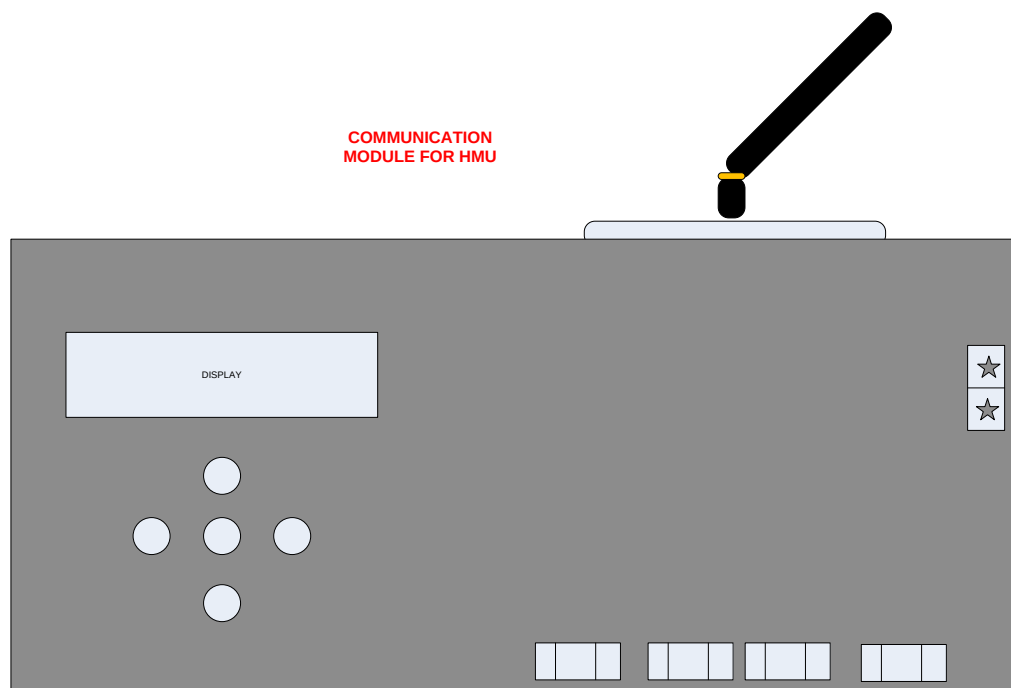
Local flash storage should be provided to each coms unit for storing the data by time stamp in case of slow data transfer .

Led indication should be provided for the identification of communication link, power indications, errors, etc.

8. Module working voltages should be of 17volts – 36 Volts DC.

Mechanical Requirements:

1. COMS UNIT should be able to enclose in the molded enclosure which can be specially designed or any other matching.



BHMU REQUIREMENT SPECIFICATION

BHMU is the battery health monitoring unit, used to measure the health of the 2volt batteries, which will be located in the battery room of the relay cabins.

Below are the battery parameters specification which needs to be monitored,

BATTERY / CHARGERS PARAMETERS		
SI No.	Item	Nominal required value
1	Float Voltage setting at battery terminals.	2.25 X No. of cells i.e., 55 cells
	(i) 24 V system	27 Volts
	(ii) 110 V system	123.75 Volts
2	Normal battery charging current when battery put into charge after discharge.	10% of AH capacity i.e. for 100 AH battery to be charged at 10 A current.
3	Maximum battery charging current when battery put in to charge after discharge.	20% of AH capacity.
4	Ripple content in DC voltage.	Should be less than 2% rms
	(i) 24 V system	0.48 V rms (Max.)
	(ii) 110 V system	2.20 V rms (Max.)
5	Battery charging voltage (High protection).	Charger should trip at 2.37 Volts per cell
	(i) For 24 Volts system	Charger should trip at 28.44 Volts
	(ii) For 110 V system	Charger should trip at 130.35 Volts.
6	Battery under voltage (Trip on discharge).	Charger should trip 1.75 Volts per cell
	(i) For 24 Volt system	Disconnect the load at 21 Volt
	(ii) For 110 Volt system	Disconnect the load at 96.25 Volt

Functional Requirements:

- Battery voltage of about 110VDC nominal should be monitored,

Integrated Power Supply (IPS) / Battery Charger:

The following shall be monitored:

- All voltage outputs
- Health of IPS modules from potential free contacts
- Availability of standby input power supplies from Auto-changeover system of

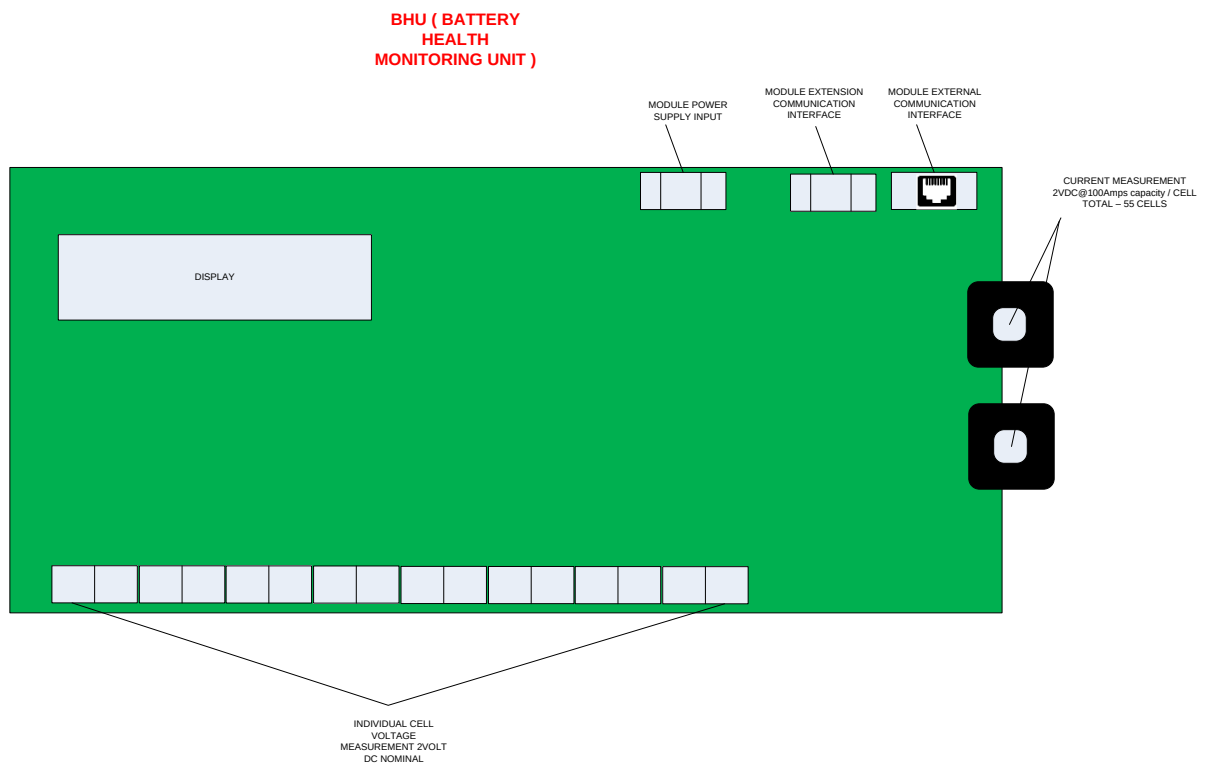
IPS.

So, the BHMU module should contains the differential circuit measurement methodology for measuring the individual cell voltage and its health.

- Each BHMU module should contain the provisions for connecting the external sensor interface for measuring the current and voltages available which is far from the place of installations.

External interface sensor connectors of about minimum, 5 channels should be provided for measuring both AC/DC voltages of about 110VDC, 230VAC.

- Non-intrusive current sensor of about 2 No's should be provided per each BHMU unit for measuring the charger current measurement of about 100 AMPS nominal., So the tolerance should be made as per the saturation voltage levels.
- Each BHMU should contain the Matrix display for displaying the value of the data measurement along with the toggle button for forwarding to the next options or value.
- Potential free contact data monitoring of about 5 No's minimum should be provided for measuring the IPS units, health.



- BHMU should be able to transmit its measured data to the COMS UNIT, through the Ethernet communications, which will be available in the Relay cabins.
- Module working voltages should be of 17volts – 36 Volts DC.

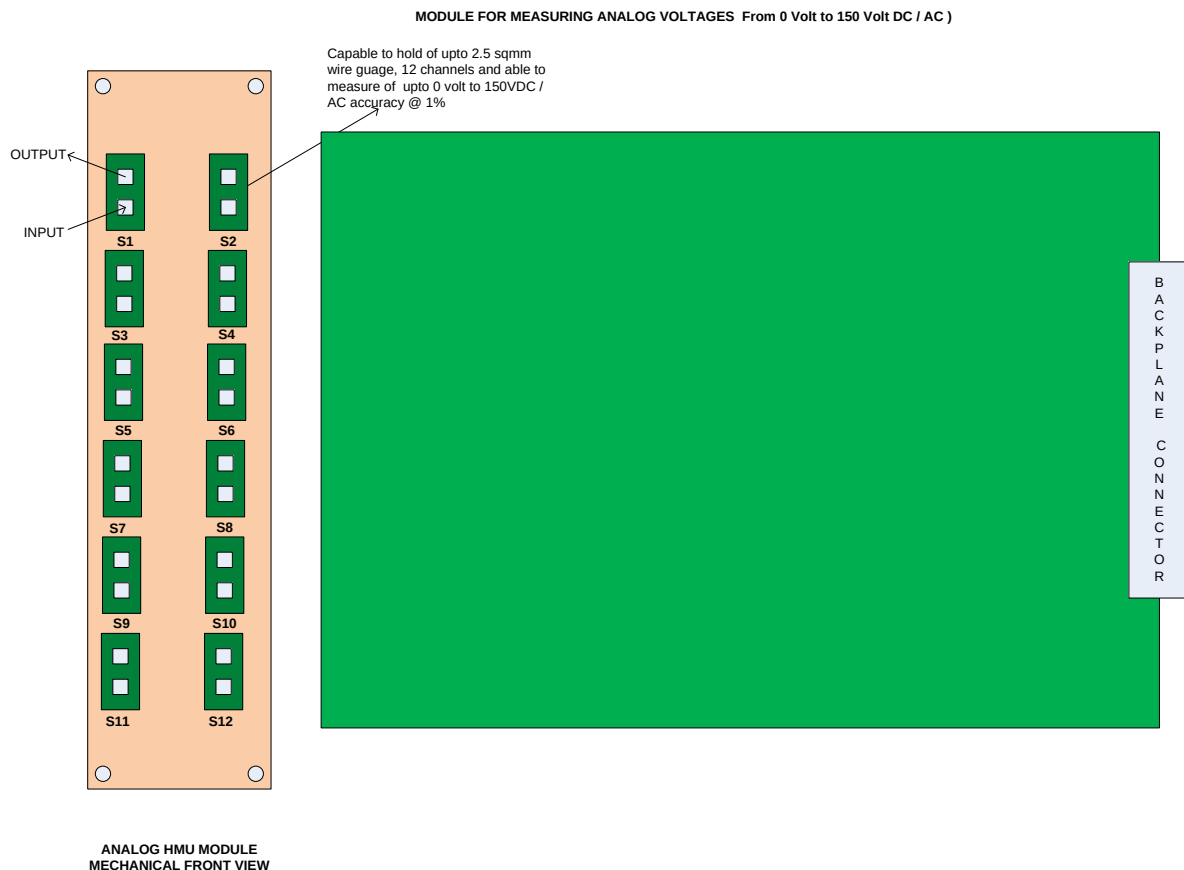
Mechanical Requirements:

- BHMU module can be able to mount it inside any of the wall mountable racks of standard size of 19' and maximum of 3U heights.

ANALOG HMU REQUIREMENT SPECIFICATION

Analog HMU module is responsible for measuring the voltages of about 0 to 150 VDC, which will be available in the Relay cabins, Channels of about 48 No's minimum is required and it should be arranged in the form of slots concepts.

ANALOG HMU



Functional Requirements:

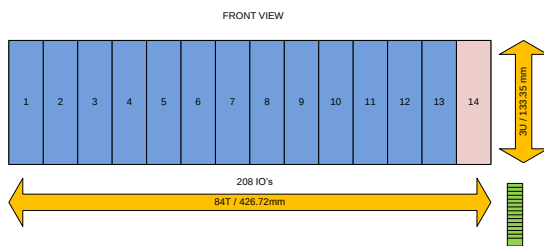
1. Each analog HMU card should be differential input with the Galvanically isolated type as per the standard of about 2.5kv Minimum, with the ports available in front of the board using phoenix connectors or similar.
2. Backplane connectors will be used for plugin the board with motherboard for channel extension to meet the requirements.

Analog HMU must have the CAN BUS/ SM BUS, communication as the communication medium, and final CAN BUS will be connected into the existing COMS UNIT (or) MASTER HMU.

3. Each port should be protected from inrush voltages and currents beyond the permitted measurement ranges.
4. Led indication of the individual channel status should be provided for the troubleshooting the gears status.
5. Analog HMU should be able to read the Temperature of the cabins, so, provision of the External Temperature sensor interface of about 2 channels should be provided with each card.

Mechanical Requirements:

1. Rack type arrangement is viable for the channel extensions for Analog HMU.

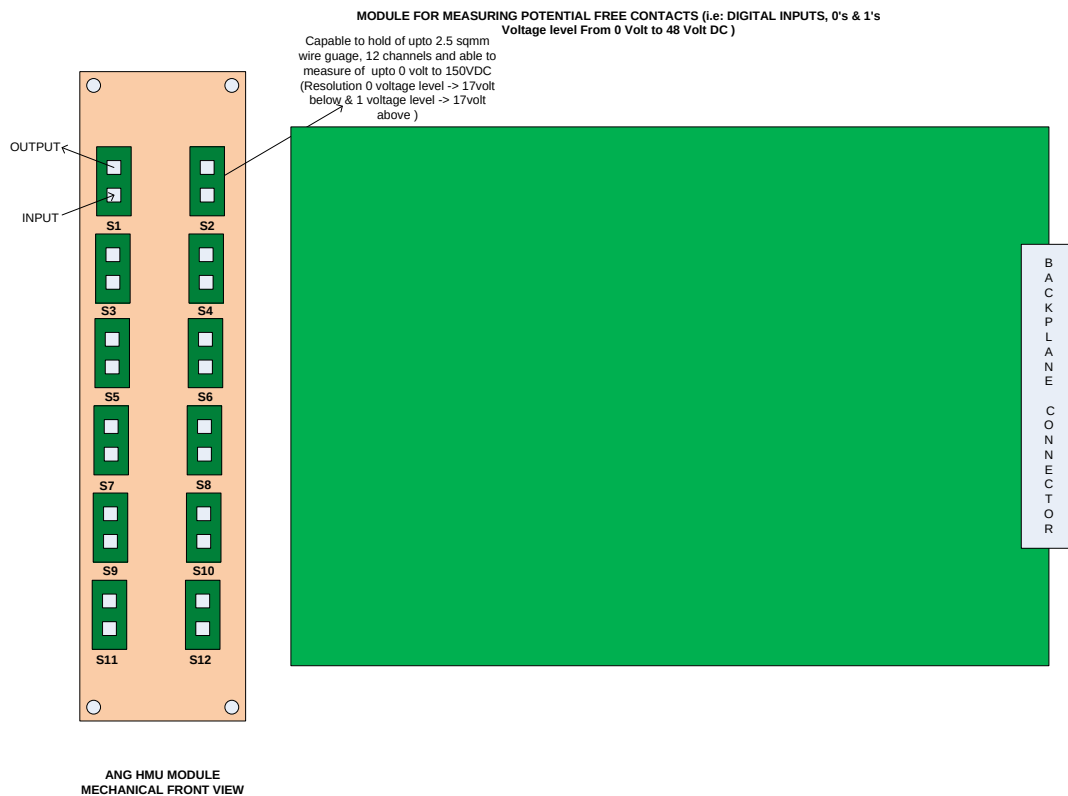


2. Size can be 3U, 84T.

Digital HMU REQUIREMENT SPECIFICATION

Digital HMU module is responsible for measuring the digital voltages of about 0 to 60 VDC, which will be available in the Relay cabins, Channels of about 512 No's.

DIGITAL HMU



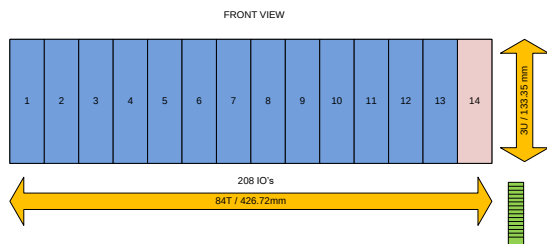
Functional Requirements:

1. Each card will sense the digital supply from the voltage limit up to 60VDC, so the necessary channel protections should be provided.
2. Each of the channels should be optically isolated may be using optocouplers for channel protection from noise and surges.
3. Digital HMU must have the CAN BUS/ SM BUS, communication as the communication medium, and final CAN BUS will be connected into the existing COMS UNIT (or) MASTER HMU.

4. Led indication of the individual channel status should be provided for the troubleshooting the gears status.

Mechanical Requirements:

1. Rack type arrangement is viable for the channel extensions for Analog HMU.



2. Size can be 3U, 84T.

MASTER HMU REQUIREMENT SPECIFICATION

Master HMU is responsible for processing the data of every HMU modules, which will be located in the central location of the station.

Below is the functional requirements, which must be implemented into the module,

Functional Requirements:

1. Ethernet port of 4 No's minimum for external data transfer to communicate with Desktop computer which will be loaded with application for data logging and offline data analyze features to be implemented.
2. Zigbee communication is required for short range data transceiver purpose from location available HMU modules using coms units through a mesh network of intermediate devices.
Data transmission range of about 80-100 meters from location to relay cabins.
3. Lora communication is required for long range data transmit from location available HMU modules to connect and communicate with relay distributed cabins coms unit.
Data transmission range of about 3 Kilometers from location to relay cabins either independently and through mesh network topologies.
Lora transmission frequency of india 865 MHz – 867 MHz bands, must be driven for license free data transceiver and options for customizing the frequency of license acquiring channel bands also can be possible in futures.
4. 2G/3G/4G/5G communication is required for communicating the COMS UNIT modules as well as for transmitting the logged station whole data to the cloud server using railnet networks.

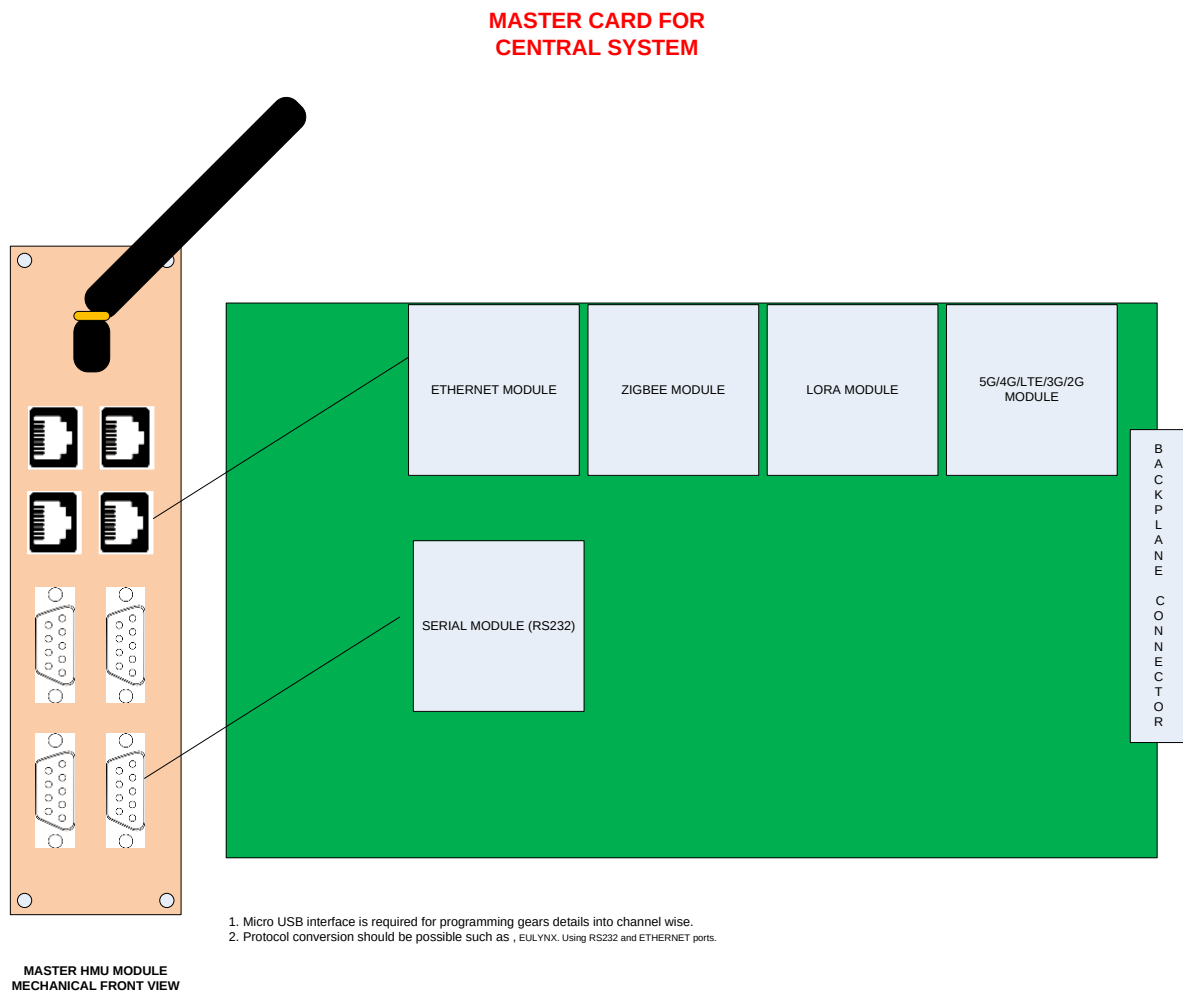
Each communication mentioned can work independently or by configuration specific one.

5. Wi-Fi communication can be provided as the additional data transmission medium.
2.5 GHz AND 5 GHz toggling options can be possible.
40 METERS.
6. CAN or SM BUS interface needs to be given for at least 4 No's should be provided for interfacing the other sensor modules such as SIGNAL HMU, POINT HMU, TCK HMU.
CAN BUS data rate of about 1 Mbps min should be provided.

Local flash storage should be provided to each coms unit for storing the data by time stamp in case of slow data transfer.

Led indication should be provided for the identification of communication link, power indications, errors, etc.

7. RS232 serial communication may be required for protocol conversion and data transmission by communicating with other OEM, subsystems.
8. Module working voltages should be of 17volts – 36 Volts DC.



9. MASTER HMU should be implemented with the following Network and Communication protocols between subsystems,

- A. The exchange of information from IoT device to Local server and to Centralized Cloud shall be on standard format for interoperability between different systems.
- B. The time synchronization is an important requirement between various IoT devices for data interpretation in machine learning. This has been taken care in the standard protocol. For master clock, the IRNSS (Indian Regional Navigation Satellite System) clock or SNTP protocol shall be taken for reference.

LOCAL SERVER AND EDGE COMPUTING REQUIREMENTS:

1. At each station there shall be centralized setup consisting of Gateway to collect the data from various IoT devices and a set of Local server with Edge computing. One integrated unit with capability to work as gateway to collect field data, edge computing etc can also be used. The basic architecture at station is depicted in Figure

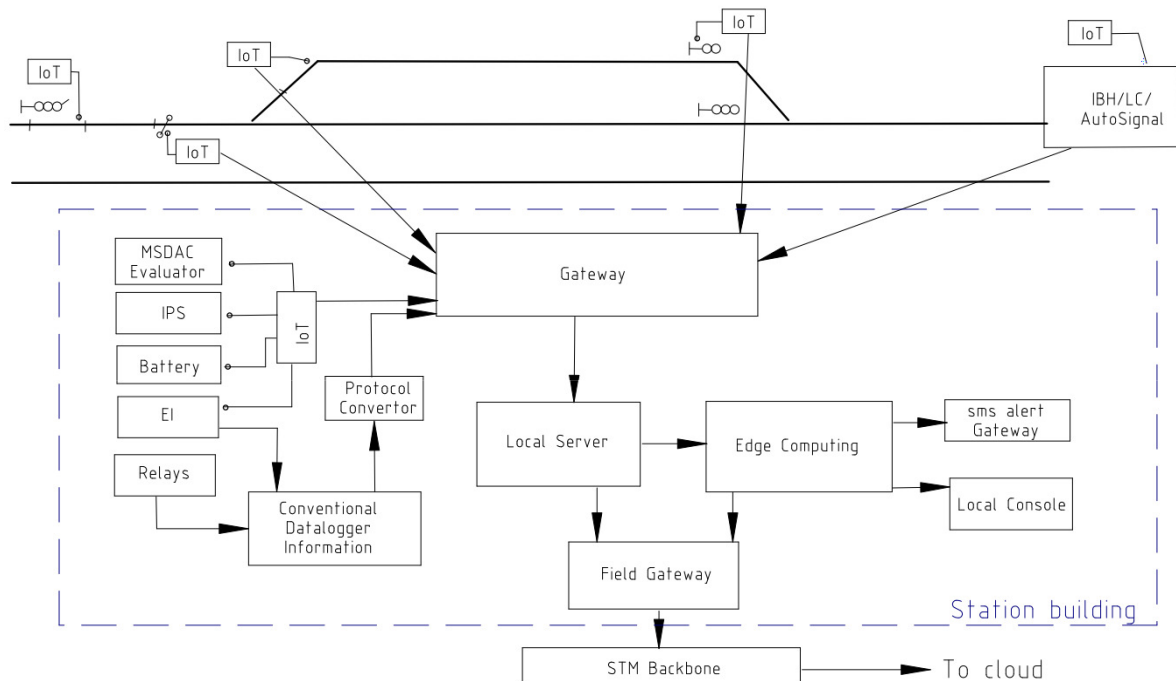


Figure-3 : Arrangement at Station

2. The Gateway shall have interface to collect data from various media like wireless, OFC and copper. It shall also be able to collect data from conventional Data Logger for indoor relays status.
3. The edge computing shall be employed to decentralize the computing and handling the big data and alert mechanism for each station.

4. Sending the alert to maintenance staff through SMS using “SMS Alert Gateway” over GPRS/LTE.
5. This alert shall also be sent to Cloud which shall be used for improving machine learning algorithms using AI techniques.

Field Gateway:

1. At each station there shall be field Gateway which filters and pre-processes the data to be sent to and received from Cloud.
2. Field gateway will transmit/receive the data to cloud over the STM backbone of RailTel. 4G/LTE communication may also be used for redundancy.

Mechanical Requirements:

1. Rack type arrangement is viable for mounting the MASTER HMU along with the digital HMU.

RDPMS SYSTEM ARCHITECTURE

